

Land–Water Boundary Detection using Custom CNN (From Scratch)

Abstract

This project presents a fully custom implementation of a Convolutional Neural Network (CNN) built from scratch (without TensorFlow/PyTorch or backpropagation) to detect land–water boundaries in satellite imagery. The system separates smooth (water-like) and chaotic (land-like) regions using handcrafted convolutional feature extraction and statistical decision logic. A Flask web application enables real-time inference and visualization.

1. Problem Statement

Given a satellite image containing any type of water body (coastline, lake, river, muddy water, blue ocean), the objective is to detect and visualize the land–water boundary without relying on pretrained deep learning models.

2. Methodology

- Preprocessing: resizing to 256x256, grayscale conversion, normalization.
- Patch-based sliding window segmentation.
- Custom CNN feature extraction (manual convolution + ReLU).
- Texture analysis using mean and variance of feature maps.
- Boundary mask generation and contour extraction.
- Red boundary overlay on original image.

3. Custom CNN Architecture

The CNN acts as a deterministic feature extractor rather than a trained classifier. Fixed convolution filters capture texture patterns, and statistical measures of feature maps are used to distinguish smooth vs chaotic regions.

4. Evaluation Strategy

Ground truth thick boundary masks were created using CVAT. A boundary coverage metric was defined as the ratio of predicted boundary pixels falling inside the ground truth tolerance band. Average Boundary Score across test set: 0.64.

5. Deployment

A Flask-based web application was developed to allow image upload, real-time processing, and boundary visualization.

6. Conclusion

This project demonstrates that meaningful land–water boundary detection can be achieved using handcrafted CNN-based feature extraction without modern deep learning frameworks. The system emphasizes interpretability, academic rigor, and engineering clarity.